What does it take to use something again?

SpaceX as a case study in reusable hardware and systems thinking

A clear question

Define what the audience needs to understand.

→

A simple model

Show the parts and how they relate.

→

Evidence

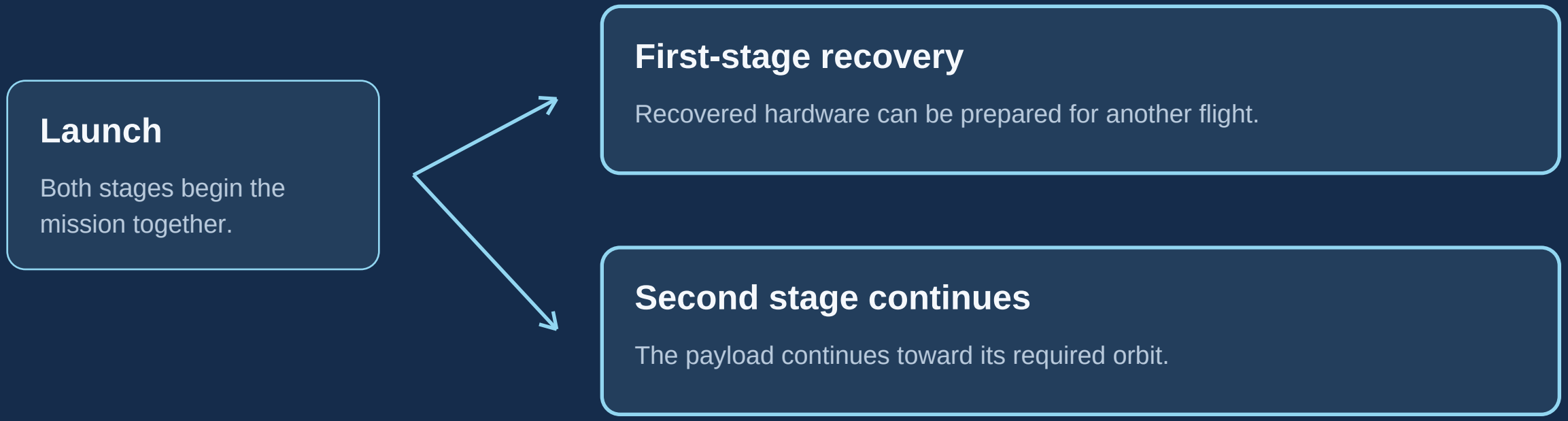
Separate facts from your interpretation.

Start with the idea. Add detail only when it helps.

Independent demo. No SpaceX or NASA affiliation. Diagrams are conceptual, not to scale.

Two stages. Different jobs.

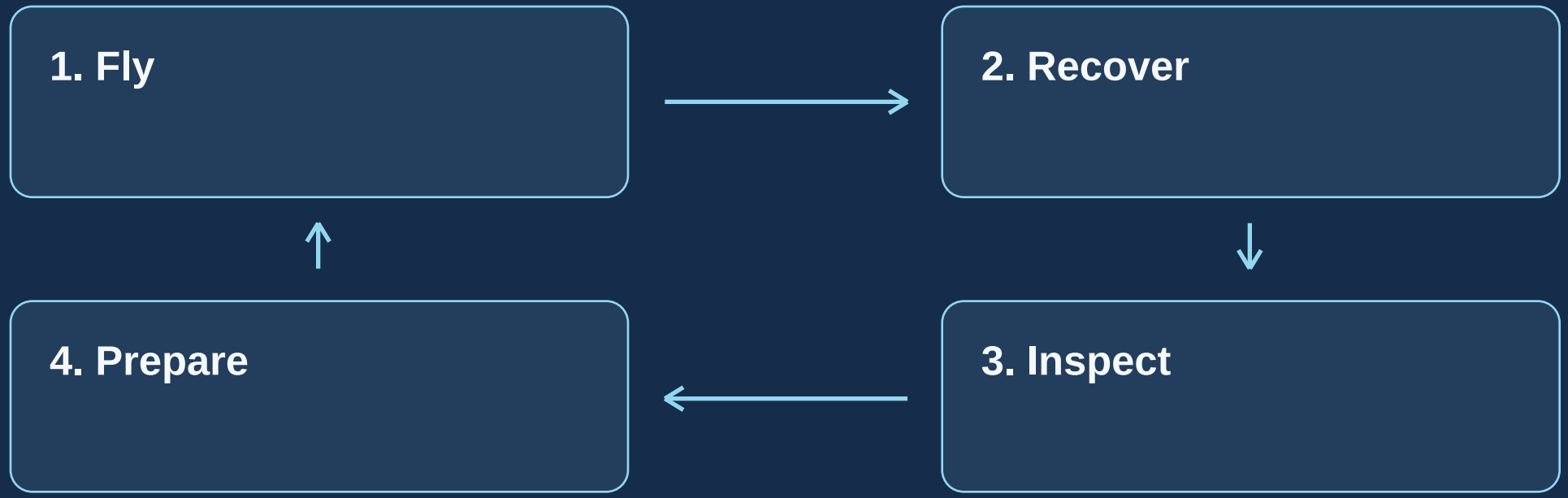
Falcon 9 divides the work of a launch between two stages.



Recovery is mission-dependent and is not guaranteed.

Recovery is part of a larger cycle.

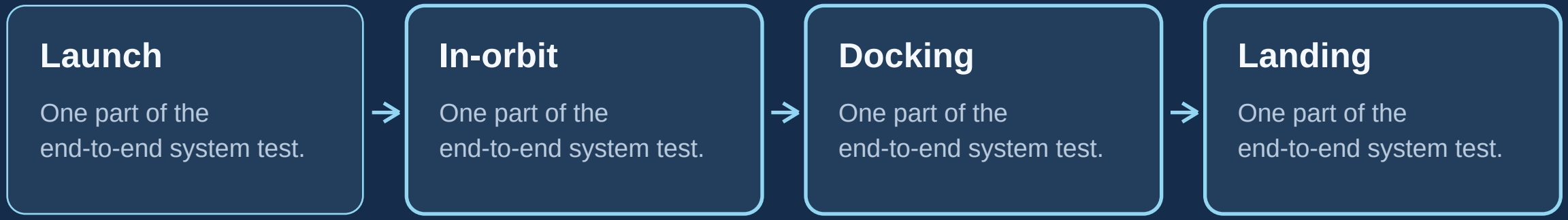
A conceptual model of reuse, not an internal company checklist.



Ask what changed, what is known, and what still needs checking.

Demo-2: test the whole journey.

May 30, 2020 / Robert Behnken and Douglas Hurley / Crew Dragon



NASA described Demo-2 as an end-to-end crew transportation test.

Explain the relationship, not every detail.

Three presentation lessons from the case study

1 State the question

What does the audience need to understand?

2 Show the handoff

How do the parts work together?

3 Separate facts and interpretation

Which claims have evidence, and which are your teaching model?

Your next explanation

Choose a system in your own work. Make its logic easier to follow.

Parts

Name the pieces that matter.

Handoffs

Show how the work moves.

Evidence

Say what is known and assumed.

Primary sources

NASA: Launch Services Program rockets (Falcon 9)

NASA: Demo-2 launch release, May 30, 2020

Full source links and accessible slide descriptions: demo/sources.html

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